

by defn. of universal quantifier, $\forall x(\neg P(x) \vee \neg R(x))$ is ~~True~~ and hence $\neg \exists x(P(x) \wedge R(x))$ is False.

∴ The conclusion does not follow $\square$

60) Let $P(x)$, $Q(x)$ and $R(x)$ be the statements "x is a clear explanation", "x is satisfactory," and "x is an excuse", respectively. Suppose that the domain for x consists of all English text. Express each of these statements using quantifiers, logical connectives and $P(x)$, $Q(x)$ and $R(x)$.

a) All clear explanations are satisfactory: $\forall x(P(x) \rightarrow Q(x))$

b) Some excuses are unsatisfactory: $\exists x(R(x) \wedge \neg Q(x))$

c) Some excuses are not clear explanations: $\exists x(R(x) \wedge \neg P(x))$

d) Does (c) follow from (a) and (b)?

Let, $\forall x(P(x) \rightarrow Q(x))$ be True. By conditional-disjunction equivalence, $\forall x(\neg P(x) \vee Q(x))$ is True. By defn. of universal quantifier, for all elements a in the domain, $\neg P(a) \vee Q(a)$ is True.

Let $\exists x(R(x) \wedge \neg Q(x))$ be True. By defn. of existential quantifier, there exists at least one element b in the domain such that $R(b) \wedge \neg Q(b)$ is True. By defn. of AND, $R(b)$ is True and $\neg Q(b)$ is True. By defn. of negation, $\neg(\neg Q(b))$ is False. By double negation law, $Q(b)$ is False.

We know, $\neg P(b) \vee Q(b)$ is True. By defn. of OR, $\neg P(b)$ must be True. By defn. of AND, $R(b) \wedge \neg P(b)$ is True. By defn. of existential quantifier, $\exists x(R(x) \wedge \neg P(x))$ is True.

∴ The conclusion (c) follows from (a) and (b) $\square$

61) Let $P(x)$, $Q(x)$, $R(x)$ and $S(x)$ be the statements "x is a baby," "x is logical," "x is able to manage a crocodile," and "x is despised," respectively.